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An Ab Initio Exciton Model Including Charge-Transfer Excited States

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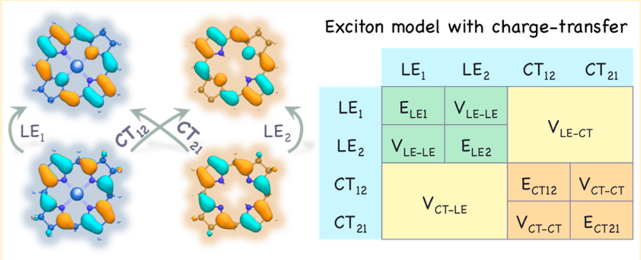
Supporting Information

ABSTRACT: The Frenkel exciton model is a useful tool for theoretical studies of multichromophore systems. We recently showed that the exciton model could be used to coarse-grain electronic structure in multichromophoric systems, focusing on singly excited exciton states [*Acc. Chem. Res.* **2014**, *47*, 2857−2866]. However, our previous implementation excluded charge-transfer excited states, which can play an important role in light-harvesting systems and near-infrared optoelectronic materials. Recent studies have also emphasized the significance of charge-transfer in singlet fission, which mediates the coupling between the locally excited states and the multiexcitonic states. In this work, we report on an ab initio exciton model that incorporates charge-transfer excited states and demonstrate that the model provides correct charge-transfer excitation energies and asymptotic behavior. Comparison with TDDFT and EOM-CC2 calculations shows that our exciton model is robust with respect to system size, screening parameter, and different density functionals. Inclusion of charge-transfer excited states makes the exciton model more useful for studies of singly excited states and provides a starting point for future construction of a model that also includes double-exciton states.



1. INTRODUCTION

The Frenkel exciton model[1−4] is a computationally advantageous scheme to describe the electronic structure of multichromophoric systems. By dividing the system into subgroups (each containing one of the constituent chromophores), it substantially decreases the computational cost for chromophoric assemblies, such as the light-harvesting system II (LH2) in purple bacteria.[5] The exciton model has been most successful in the weak coupling limit, where the excitons are of Frenkel-type, i.e., localized on each individual chromophore. This introduces the assumption that the electrons are confined within each chromophoric unit, whereas in reality electron transfer between chromophore units may occur after optical excitation. In photosynthesis[6,7] and solar cells,[8] such electron transfer or charge transfer[9,10] is an important process that converts photon energy into chemical energy or electric current. Charge-transfer excited states also play important roles in near-infrared absorption[11] as well as in singlet fission[12,13] and thus can be important for efficient harvesting of solar energy. The development of charge-transfer complexes and materials inevitably demands fundamental understanding of the underlying excitations and electronic transitions.

Quantum chemistry provides practical approaches for studies on excited states. On one hand, high-level wave function-based methods, for instance, EOM-CC2[14,15] and CASPT2,[16] are capable of accurately describing valence, Rydberg, and charge-transfer type excitations, but their computational cost increases sharply with respect to system size, which often makes the calculations prohibitive. On the other hand, the computationally efficient time-dependent density functional theory (TDDFT) approach[17] has practical deficiencies for charge-transfer excited states[18−20] and, in the usual linear response formalism, also has difficulty with doubly excited states[21] and conical intersections.[22] The major source of error in TDDFT is the exchange-correlation kernel (K_{xc}). Most modern approximations to K_{xc} decay too quickly and fail to describe the $1/R$ asymptotic behavior of the Coulombic attraction between the hole and the electron.[19] The introduction of long-range corrected (LRC) hybrid density functionals (e.g., LRC-ωPBEh) alleviates the problem,[23] but the result remains sensitive to the chosen density functional as well as the screening parameter ω. Kronik and co-workers proposed an optimal-tuning approach,[24,25] where the screening parameter is chosen to satisfy Koopmans' theorem for the ionization potential. Optimal-tuning TDDFT has been shown to work nicely in a number of systems, but it can be challenging to find a universal screening parameter for a heterogeneous system. Indeed, for extended conjugated molecules, it has been shown that the optimal screening parameter judged by ionization potential has an opposite size dependence compared to the optimal screening parameter judged by vertical excitation energy (compared to the experimental absorption maximum).[26]

Many-electron self-interaction error in approximate density functionals is a major source of intrinsic errors in DFT and TDDFT.[27] These errors arise from the delocalization error of semilocal density functionals,[28] where electrons are overly delocalized because the energy for fractionally occupied states is

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convex and nonlinear with respect to electron occupation.[28−31] Hartree−Fock calculations, on the contrary, suffer from localization error because the energy of fractionally occupied states is concave and nonlinear with respect to electron occupation.[28] Mixing of Hartree−Fock exchange with density functionals improves the performance in certain cases but does not guarantee a cancellation of errors, and the problem persists in LRC hybrid functionals. Two efforts to overcome the problem involve the recently introduced local scaling correction (LSC)[32] and constrained density functional theory (CDFT).[33,34] The LSC approach offers promising performance for molecular dissociation, transition states, and charge-transfer systems, whereas CDFT provides a direct construction of diabatic electronic states that are naturally suitable to address charge-transfer. In addition, the combination of CDFT with configuration interaction (CI) allows multiple diabatic states to be mixed through the construction of a physically motivated effective Hamiltonian.[35−37]

The above-mentioned optimal-tuning TDDFT, LSC-DFT, and CDFT-CI approaches are based on the self-consistent field (SCF) procedure employed in standard Hartree−Fock or DFT calculations. The computational cost normally scales as $O(N^3)$, where N denotes the number of basis functions in the system. This prohibits direct studies on very large systems such as the pigment assembly in the LH2 system of purple bacteria,[38] which consists of 27 bacteriochlorophyll and 9 carotenoid molecules. A scalable approach that can tackle excited state calculations including both valence and charge-transfer excitations is desirable for theoretical studies. Earlier efforts involve the collective optical excitation approach,[39] effective Hamiltonian based on Förster theory,[40] vibration-coupled effective mode analysis,[41] etc. We have previously addressed this problem for intramonomer valence excitations by developing an ab initio exciton model which constructs a Frenkel exciton Hamiltonian on the fly using monomer quantum chemistry calculations.[42] An approach in the same spirit, but somewhat different in details, was subsequently suggested by others.[43,44] In the present work, we extend our ab initio exciton model to include both intramonomer valence excitations and intermonomer charge transfer excitations.

2. THEORY

Our expanded Frenkel exciton model adopts the following matrix form of the Hamiltonian[45]

$$\mathbf{H} = \sum_{I}^{N} E_I |\varphi_I\rangle\langle\varphi_I| + \sum_{J\neq I}^{N} V_{IJ}|\varphi_I\rangle\langle\varphi_J| \tag{1}$$

where N is the total number of excited states in the chromophore assembly, and φ_I denotes excitonic basis function of state I. Note that, however, in the original Frenkel exciton model, the summation runs over all monomers since the excitons reside fully in monomers. Our expanded exciton model adopts a more generalized representation where the summation runs over all diabatic excited states, which are not necessarily confined within monomers. The detailed structure of the exciton model Hamiltonian matrix will be elaborated in the following sections, and applications of the exciton model will be demonstrated in three dimeric systems and one multichromophoric system.

2.1. Excitonic Basis Functions.

In the exciton model, the ground state of the system is described by an antisymmetrized product of monomer wave functions. To make exciton model calculations practical for large systems, three approximations are introduced. First, the individual monomeric wave functions are approximated by the wave functions of the isolated monomers (which we call the frozen density approximation below). Second, molecular orbitals from different monomers are approximated to be orthogonal in deriving matrix element expressions. This is not strictly correct, since the molecular orbitals of each monomer are determined independently. However, the intermolecular overlap typically decreases very quickly with the distance between monomers, making it a viable approximation in many real-life systems. Third, the electron repulsion integrals involving orbitals located on three or four monomers are neglected. In practice, it is often the case that monomers are sufficiently distant to justify these three approximations for the exciton model. The ground state wave function of the system thus becomes

$$|\Psi_0\rangle = \hat{\mathcal{A}}|\Psi_A\Psi_B\cdots\Psi_N\rangle \approx \hat{\mathcal{A}}|\Psi_A^{(0)}\Psi_B^{(0)}\cdots\Psi_N^{(0)}\rangle \tag{2}$$

where $\hat{\mathcal{A}}$ is the antisymmetrization operator and $\Psi_A^{(0)}$ is the wave function for the isolated monomer A.

Two types of excited states are involved in the exciton model developed herein, namely the locally excited (LE) states and the charge-transfer (CT) excited states. Only singlet excited states are considered in this work. An LE state corresponds to an electronic excitation located on a single monomer. The wave function for an LE state on monomer A is

$$\begin{aligned}|\Psi_{LE}^{A(n)}\rangle &= \sum_{i_A a_A} c_{i_A a_A}^{(n)}|\Psi_{i_A}^{a_A}\rangle \\ &= \frac{1}{\sqrt{2}}\sum_{i_A a_A} c_{i_A a_A}^{(n)}(\{\bar{a}_A^\dagger \bar{i}_A\} + \{a_A^\dagger i_A\})|\Psi_0\rangle\end{aligned} \tag{3}$$

where $c_{i_A a_A}^{(n)}$ denotes the CI coefficient of the $i_A \to a_A$ transition in the nth LE state of monomer A, i_A/a_A denote annihilation operators corresponding to occupied/virtual orbitals localized on monomer A, and the presence/absence of an overbar denotes β/α spin. A CT state corresponds to excitation of an electron across monomers, and the wave function for a CT state can be written in a similar manner to eq 3:

$$\begin{aligned}|\Psi_{CT}^{A\to B}\rangle &= \sum_{i_A a_B} c_{i_A a_B}|\Psi_{i_A}^{a_B}\rangle \\ &= \frac{1}{\sqrt{2}}\sum_{i_A a_B} c_{i_A a_B}(\{\bar{a}_B^\dagger \bar{i}_A\} + \{a_B^\dagger i_A\})|\Psi_0\rangle\end{aligned} \tag{4}$$

In this work we restrict the CT state to electronic excitation between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), such that the CT state corresponds to an integer CT from one monomer to another. The wave function for the CT state in eq 4 becomes

$$|\Psi_{CT}^{A\to B}\rangle = \frac{1}{\sqrt{2}}(\{\bar{l}_B^\dagger \bar{h}_A\} + \{l_B^\dagger h_A\})|\Psi_0\rangle \tag{5}$$

where h_A and l_B denote the HOMO of monomer A and the LUMO of monomer B, respectively.

Our exciton model involves three types of states, the ground state (GS), the LE states, and the CT states. Since we used the frozen density approximation, the GS−LE and GS−CT couplings do not vanish (these would vanish by Brillouin's theorem if the fully variational molecular orbitals of the supersystem were employed). However, these couplings introduce a size-extensivity problem analogous to that found

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in excitation-level truncated CI methods (see the Supporting Information for further discussion). We avoid this problem by discarding the GS−LE and GS−CT couplings, constructing the exciton model Hamiltonian matrix using only LE and CT states as basis functions. This implies that the model focuses on excited states and will not be able to describe conical intersections between the ground and lowest excited adiabatic states.[22] Thus, the diagonal elements of the Hamiltonian matrix correspond to the excitation energies of the LE and CT diabatic states, and the off-diagonal elements correspond to the couplings among these states. In the next two sections, we provide expressions for the diagonal and off-diagonal matrix elements of the Hamiltonian.

2.2. Excitation Energies. The excitation energy of the nth LE state on monomer A is

$$\begin{aligned} E_{\mathrm{LE}}^{\mathrm{A}(n)} &= \langle \Psi_{\mathrm{LE}}^{\mathrm{A}(n)} | \hat{\mathcal{H}} | \Psi_{\mathrm{LE}}^{\mathrm{A}(n)} \rangle - \langle \Psi_0 | \hat{\mathcal{H}} | \Psi_0 \rangle \\ &\approx \sum_{i_\mathrm{A} a_\mathrm{A}} \sum_{j_\mathrm{A} b_\mathrm{A}} c_{i_\mathrm{A} a_\mathrm{A}}^{(n)} c_{j_\mathrm{A} b_\mathrm{A}}^{(n)} [\delta_{i_\mathrm{A} j_\mathrm{A}} f_{a_\mathrm{A} b_\mathrm{A}} - \delta_{a_\mathrm{A} b_\mathrm{A}} f_{i_\mathrm{A} j_\mathrm{A}} + 2(i_A a_A | j_A b_A) \\ &\quad - c_{\mathrm{HF}} (i_A j_A | a_A b_A) + (1 - c_{\mathrm{HF}})(i_A a_A | f_{\mathrm{xc}} | j_A b_A)] \\ &= E_{\mathrm{LE}(0)}^{\mathrm{A}(n)} + \sum_{i_\mathrm{A} a_\mathrm{A}} \sum_{j_\mathrm{A} b_\mathrm{A}} c_{i_\mathrm{A} a_\mathrm{A}}^{(n)} c_{j_\mathrm{A} b_\mathrm{A}}^{(n)} [\delta_{i_\mathrm{A} j_\mathrm{A}} f_{a_\mathrm{A} b_\mathrm{A}}^{\Delta} - \delta_{a_\mathrm{A} b_\mathrm{A}} f_{i_\mathrm{A} j_\mathrm{A}}^{\Delta}] \end{aligned} \tag{6}$$

where c_{HF} is the coefficient of Hartree−Fock exchange, f_{xc} is the employed exchange-correlation functional, and f_{ab} and f_{ij} are the Fock matrix elements of monomer A in the presence of other monomers. In practice, we use TDDFT to calculate the excitation energy of monomer A, $E_{\mathrm{LE}(0)}^{\mathrm{A}(n)}$, as well as the CI coefficients, $c_{i_\mathrm{A} a_\mathrm{A}}^{(n)}$. The electrostatic contribution to the Fock matrix due to other monomers is approximated as (in the atomic orbital basis):

$$f_{\mu\nu}^{\Delta} = -\sum_{\mathrm{B} \neq \mathrm{A}} \sum_{k \in \mathrm{B}} q_k^{\mathrm{B}} \int \mathrm{d}\vec{r} \frac{\phi_\mu(\vec{r}) \phi_\nu(\vec{r})}{|\vec{r} - \vec{R}_k^{\mathrm{B}}|} \tag{7}$$

where the remaining monomers are represented with an atomic point charge decomposition (q_k^{B} is the atomic charge for the kth atom in the B monomer). Here, the atomic point charges are obtained by fitting to the electrostatic potential (ESP) on grid points surrounding the molecule. The details of this ESP fitting are described in the Supporting Information.

For a CT state, it is possible to evaluate the excitation energy using a similar formalism to that in eq 6

$$\begin{aligned} E_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} &= \langle \Psi_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} | \hat{\mathcal{H}} | \Psi_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} \rangle - \langle \Psi_0 | \hat{\mathcal{H}} | \Psi_0 \rangle \\ &\approx f_{l_\mathrm{B} l_\mathrm{B}} - f_{h_\mathrm{A} h_\mathrm{A}} + 2(h_\mathrm{A} l_\mathrm{B} | h_\mathrm{A} l_\mathrm{B}) - c_{\mathrm{HF}} (h_\mathrm{A} h_\mathrm{A} | l_\mathrm{B} l_\mathrm{B}) \\ &\quad + (1 - c_{\mathrm{HF}})(h_\mathrm{A} l_\mathrm{B} | f_{\mathrm{xc}} | h_\mathrm{A} l_\mathrm{B}) \end{aligned} \tag{8}$$

However, it is known that TDDFT suffers from inherent self-interaction errors as well as incorrect asymptotic behavior of CT excitation energies when pure or global hybrid density functionals are employed. Therefore, instead of using TDDFT for this purpose, we employ a ΔSCF-like[46,47] density-matrix-based approach to calculate the CT excitation energy

$$\begin{aligned} E_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} &= \langle \Psi_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} | \hat{\mathcal{H}} | \Psi_{\mathrm{CT}}^{\mathrm{A} \to \mathrm{B}} \rangle - \langle \Psi_0 | \hat{\mathcal{H}} | \Psi_0 \rangle \\ &\approx E(\mathrm{A^+B^-}) - E(\mathrm{AB}) \\ &= \frac{1}{2} \sum_{pq\sigma} [D_{pq,\sigma}^{\mathrm{A^+B^-}} (h_{pq}^{\mathrm{A^+B^-}} + \tilde{f}_{pq,\sigma}^{\mathrm{A^+B^-}}) \\ &\quad - D_{pq,\sigma}^{\mathrm{AB}} (h_{pq}^{\mathrm{AB}} + \tilde{f}_{pq,\sigma}^{\mathrm{AB}})] + E_{\mathrm{xc}}^{\mathrm{A^+B^-}} - E_{\mathrm{xc}}^{\mathrm{AB}} \end{aligned} \tag{9}$$

where p and q are atomic orbitals, σ denotes the spin of electrons (α or β), D_{pq} and h_{pq} are respectively density matrix elements and core Hamiltonian matrix elements, $\tilde{f}_{pq}$ is the Fock matrix element without exchange−correlation contribution, and E_{xc} is the exchange−correlation energy. Here, based on the aforementioned frozen density approximation, the density matrix **D** for the neutral dimer AB is constructed as a block-diagonal matrix consisting of density matrices of the neutral monomers A and B. Similarly, the **D** matrix of the charge-transfer dimer $\mathrm{A^+B^-}$ is constructed from the density matrices of the ionic monomers $\mathrm{A^+}$ and $\mathrm{B^-}$. The Fock matrix for the neutral or charge-transfer dimer is then computed from the appropriate **D**. When there are more than two chromophores in the system, the electrostatic effects of the ESP point charges of chromophores other than A and B are included in the Fock matrix through the nuclear attraction integral in analogy to eq 7. In DFT calculations, the exchange-correlation contribution is also included in the CT excitation energy. As such, the ionization potential of the donor, the electron affinity of the acceptor, and the Coulomb attraction between the hole and the electron are all taken into account, and the result is much less prone to self-interaction errors compared to TDDFT.

2.3. Couplings. The coupling between two LE states on the same monomer is calculated as

$$\begin{aligned} V_{\mathrm{LE-LE}}^{\mathrm{A}(m),\mathrm{A}(n)} &= \langle \Psi_{\mathrm{LE}}^{\mathrm{A}(m)} | \hat{\mathcal{H}} | \Psi_{\mathrm{LE}}^{\mathrm{A}(n)} \rangle \\ &\approx \sum_{i_\mathrm{A} a_\mathrm{A}} \sum_{j_\mathrm{A} b_\mathrm{A}} c_{i_\mathrm{A} a_\mathrm{A}}^{(m)} c_{j_\mathrm{A} b_\mathrm{A}}^{(n)} [\delta_{i_\mathrm{A} j_\mathrm{A}} f_{a_\mathrm{A} b_\mathrm{A}} - \delta_{a_\mathrm{A} b_\mathrm{A}} f_{i_\mathrm{A} j_\mathrm{A}} \\ &\quad + 2(i_\mathrm{A} a_\mathrm{A} | j_\mathrm{A} b_\mathrm{A}) - c_{\mathrm{HF}} (i_\mathrm{A} j_\mathrm{A} | a_\mathrm{A} b_\mathrm{A}) \\ &\quad + (1 - c_{\mathrm{HF}})(i_\mathrm{A} a_\mathrm{A} | f_{\mathrm{xc}} | j_\mathrm{A} b_\mathrm{A})] \\ &= \sum_{i_\mathrm{A} a_\mathrm{A}} \sum_{j_\mathrm{A} b_\mathrm{A}} c_{i_\mathrm{A} a_\mathrm{A}}^{(m)} c_{j_\mathrm{A} b_\mathrm{A}}^{(n)} [\delta_{i_\mathrm{A} j_\mathrm{A}} f_{a_\mathrm{A} b_\mathrm{A}}^{\Delta} - \delta_{a_\mathrm{A} b_\mathrm{A}} f_{i_\mathrm{A} j_\mathrm{A}}^{\Delta}] \end{aligned} \tag{10}$$

where $m \neq n$. For an isolated monomer A, $V_{\mathrm{LE-LE}}^{\mathrm{A}(m),\mathrm{A}(n)} = 0$. However, in the presence of other monomers, $V_{\mathrm{LE-LE}}^{\mathrm{A}(m),\mathrm{A}(n)}$ may deviate from zero due to perturbations to the Fock matrix elements (denoted as f_{ab}^{Δ} and f_{ij}^{Δ}) arising from the electrostatic influence of surrounding monomers.

The coupling between two LE states on different monomers has only two-electron contributions:

$$\begin{aligned} V_{\mathrm{LE-LE}}^{\mathrm{A}(m),\mathrm{B}(n)} &= \langle \Psi_{\mathrm{LE}}^{\mathrm{A}(m)} | \hat{\mathcal{H}} | \Psi_{\mathrm{LE}}^{\mathrm{B}(n)} \rangle \\ &\approx \sum_{i_\mathrm{A} a_\mathrm{A}} \sum_{j_\mathrm{B} b_\mathrm{B}} c_{i_\mathrm{A} a_\mathrm{A}}^{(m)} c_{j_\mathrm{B} b_\mathrm{B}}^{(n)} [2(i_\mathrm{A} a_\mathrm{A} | j_\mathrm{B} b_\mathrm{B}) - c_{\mathrm{HF}} (i_\mathrm{A} j_\mathrm{B} | a_\mathrm{A} b_\mathrm{B}) \\ &\quad + (1 - c_{\mathrm{HF}})(i_\mathrm{A} a_\mathrm{A} | f_{\mathrm{xc}} | j_\mathrm{B} b_\mathrm{B})] \end{aligned} \tag{11}$$

Similarly, the coupling between an LE state and a CT state is

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Table 1. Structure of an Example of the Exciton Model Hamiltonian Matrix for a Dimer System Consisting of Monomers A and B[a]

	$\vert\Psi_{\mathrm{LE}}^{\mathrm{A}(1)}\rangle$	$\vert\Psi_{\mathrm{LE}}^{\mathrm{A}(2)}\rangle$	$\vert\Psi_{\mathrm{LE}}^{\mathrm{B}(1)}\rangle$	$\vert\Psi_{\mathrm{LE}}^{\mathrm{B}(2)}\rangle$	$\vert\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}\rangle$	$\vert\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{A}}\rangle$
$\langle\Psi_{\mathrm{LE}}^{\mathrm{A}(1)}\vert$	$E_{\mathrm{LE}}^{\mathrm{A}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(1),\mathrm{A}(2)}$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(1),\mathrm{B}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(1),\mathrm{B}(2)}$	$V_{\mathrm{LE-CT}}^{\mathrm{A}(1),\mathrm{A}\to\mathrm{B}}$	$V_{\mathrm{LE-CT}}^{\mathrm{A}(1),\mathrm{B}\to\mathrm{A}}$
$\langle\Psi_{\mathrm{LE}}^{\mathrm{A}(2)}\vert$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(2),\mathrm{A}(1)}$	$E_{\mathrm{LE}}^{\mathrm{A}(2)}$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(2),\mathrm{B}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{A}(2),\mathrm{B}(2)}$	$V_{\mathrm{LE-CT}}^{\mathrm{A}(2),\mathrm{A}\to\mathrm{B}}$	$V_{\mathrm{LE-CT}}^{\mathrm{A}(2),\mathrm{B}\to\mathrm{A}}$
$\langle\Psi_{\mathrm{LE}}^{\mathrm{B}(1)}\vert$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(1),\mathrm{A}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(1),\mathrm{A}(2)}$	$E_{\mathrm{LE}}^{\mathrm{B}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(1),\mathrm{B}(2)}$	$V_{\mathrm{LE-CT}}^{\mathrm{B}(1),\mathrm{A}\to\mathrm{B}}$	$V_{\mathrm{LE-CT}}^{\mathrm{B}(1),\mathrm{B}\to\mathrm{A}}$
$\langle\Psi_{\mathrm{LE}}^{\mathrm{B}(2)}\vert$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(2),\mathrm{A}(1)}$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(2),\mathrm{A}(2)}$	$V_{\mathrm{LE-LE}}^{\mathrm{B}(2),\mathrm{B}(1)}$	$E_{\mathrm{LE}}^{\mathrm{B}(2)}$	$V_{\mathrm{LE-CT}}^{\mathrm{B}(2),\mathrm{A}\to\mathrm{B}}$	$V_{\mathrm{LE-CT}}^{\mathrm{B}(2),\mathrm{B}\to\mathrm{A}}$
$\langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}\vert$	$V_{\mathrm{CT-LE}}^{\mathrm{A}\to\mathrm{B},\mathrm{A}(1)}$	$V_{\mathrm{CT-LE}}^{\mathrm{A}\to\mathrm{B},\mathrm{A}(2)}$	$V_{\mathrm{CT-LE}}^{\mathrm{A}\to\mathrm{B},\mathrm{B}(1)}$	$V_{\mathrm{CT-LE}}^{\mathrm{A}\to\mathrm{B},\mathrm{B}(2)}$	$E_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}$	$V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{B},\mathrm{B}\to\mathrm{A}}$
$\langle\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{A}}\vert$	$V_{\mathrm{CT-LE}}^{\mathrm{B}\to\mathrm{A},\mathrm{A}(1)}$	$V_{\mathrm{CT-LE}}^{\mathrm{B}\to\mathrm{A},\mathrm{A}(2)}$	$V_{\mathrm{CT-LE}}^{\mathrm{B}\to\mathrm{A},\mathrm{B}(1)}$	$V_{\mathrm{CT-LE}}^{\mathrm{B}\to\mathrm{A},\mathrm{B}(2)}$	$V_{\mathrm{CT-CT}}^{\mathrm{B}\to\mathrm{A},\mathrm{A}\to\mathrm{B}}$	$E_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{A}}$

[a]The basis functions of the exciton model include two LE states on each monomer and two CT states on the dimer.

$$\begin{aligned}V_{\mathrm{LE-CT}}^{\mathrm{A}(n),\mathrm{A}\to\mathrm{B}} &= \langle\Psi_{\mathrm{LE}}^{\mathrm{A}(n)}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}\rangle\\ &\approx \sum_{i_\mathrm{A}a_\mathrm{A}} c_{i_\mathrm{A}a_\mathrm{A}}^{(n)}[\delta_{i_\mathrm{A}h_\mathrm{A}}f_{a_\mathrm{A}l_\mathrm{B}} + 2(i_\mathrm{A}a_\mathrm{A}|h_\mathrm{A}l_\mathrm{B})\\ &\quad - c_{\mathrm{HF}}(i_\mathrm{A}h_\mathrm{A}|a_\mathrm{A}l_\mathrm{B}) + (1-c_{\mathrm{HF}})(i_\mathrm{A}a_\mathrm{A}|f_{\mathrm{xc}}|h_\mathrm{A}l_\mathrm{B})]\end{aligned} \quad (12)$$

$$\begin{aligned}V_{\mathrm{LE-CT}}^{\mathrm{A}(n),\mathrm{B}\to\mathrm{A}} &= \langle\Psi_{\mathrm{LE}}^{\mathrm{A}(n)}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{A}}\rangle\\ &\approx \sum_{i_\mathrm{A}a_\mathrm{A}} c_{i_\mathrm{A}a_\mathrm{A}}^{(n)}[-\delta_{a_\mathrm{A}l_\mathrm{A}}f_{i_\mathrm{A}h_\mathrm{B}} + 2(i_\mathrm{A}a_\mathrm{A}|h_\mathrm{B}l_\mathrm{A})\\ &\quad - c_{\mathrm{HF}}(i_\mathrm{A}h_\mathrm{B}|a_\mathrm{A}l_\mathrm{A}) + (1-c_{\mathrm{HF}})(i_\mathrm{A}a_\mathrm{A}|f_{\mathrm{xc}}|h_\mathrm{B}l_\mathrm{A})]\end{aligned} \quad (13)$$

where the leading contribution comes from the one-electron terms (Fock matrix elements). If the LE state and the CT state have no common monomer, the coupling vanishes:

$$\begin{aligned}V_{\mathrm{LE-CT}}^{\mathrm{A}(n),\mathrm{B}\to\mathrm{C}} &= \langle\Psi_{\mathrm{LE}}^{\mathrm{A}(n)}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{C}}\rangle \approx \sum_{i_\mathrm{A}a_\mathrm{A}} c_{i_\mathrm{A}a_\mathrm{A}}^{(n)}[2(i_\mathrm{A}a_\mathrm{A}|h_\mathrm{B}l_\mathrm{C})\\ &- c_{\mathrm{HF}}(i_\mathrm{A}h_\mathrm{B}|a_\mathrm{A}l_\mathrm{C}) + (1-c_{\mathrm{HF}})(i_\mathrm{A}a_\mathrm{A}|f_{\mathrm{xc}}|h_\mathrm{B}l_\mathrm{C})] \approx 0\end{aligned} \quad (14)$$

because electron repulsion integrals involving orbitals located on three different monomers are neglected.

Finally, the couplings between two CT states are given as

$$\begin{aligned}V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{B},\mathrm{B}\to\mathrm{A}} &= \langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{A}}\rangle\\ &\approx 2(h_\mathrm{A}l_\mathrm{B}|h_\mathrm{B}l_\mathrm{A}) - c_{\mathrm{HF}}(h_\mathrm{A}h_\mathrm{B}|l_\mathrm{B}l_\mathrm{A})\\ &\quad + (1-c_{\mathrm{HF}})(h_\mathrm{A}l_\mathrm{B}|f_{\mathrm{xc}}|h_\mathrm{B}l_\mathrm{A})\end{aligned} \quad (15)$$

$$V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{C},\mathrm{B}\to\mathrm{C}} = \langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{C}}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{C}}\rangle \approx -f_{h_\mathrm{A}h_\mathrm{B}} \quad (16)$$

$$V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{B},\mathrm{A}\to\mathrm{C}} = \langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{C}}\rangle \approx f_{l_\mathrm{B}l_\mathrm{C}} \quad (17)$$

$$V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{B},\mathrm{B}\to\mathrm{C}} = \langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{B}\to\mathrm{C}}\rangle \approx 0 \quad (18)$$

$$V_{\mathrm{CT-CT}}^{\mathrm{A}\to\mathrm{B},\mathrm{C}\to\mathrm{D}} = \langle\Psi_{\mathrm{CT}}^{\mathrm{A}\to\mathrm{B}}|\hat{\mathcal{H}}|\Psi_{\mathrm{CT}}^{\mathrm{C}\to\mathrm{D}}\rangle \approx 0 \quad (19)$$

where we again neglect the electron repulsion integrals involving orbitals located on more than two different monomers.

2.4. Computational Procedure. Table 1 shows an example of the exciton model Hamiltonian matrix for a dimer with two LE states on each monomer and two CT states on the dimer. The Hamiltonian matrix can be straightforwardly expanded to include more LE states and multiple chromophores using the aforementioned formulas for the energies and couplings.

In the exciton model, the basis functions consist of LE and CT states. For a chromophore monomer, one or more LE states can be involved. For a chromophore pair consisting of monomers A and B, two CT states of opposite directions (A → B and B → A) are involved, corresponding to interchromophore CT excitation from the HOMO of A to the LUMO of B, and from the HOMO of B to the LUMO of A, respectively. As such, for a multichromophoric system consisting of N units, the number of LE states is proportional to N, and the number of CT states is $N(N-1)$. The total number of states in the exciton model is thus $N(N-1+M)$ where M is the number of LE states on each chromophore monomer.

We follow the procedure shown in Scheme 1 to compute all the necessary elements for the exciton model Hamiltonian matrix, using the aforementioned formulas for excitation energies and couplings. The first step is to loop over each chromophore monomer and calculate the ground state, excited states, and ionic states (cation and anion) in vacuum, for which

Scheme 1. Computational Procedure for the Exciton Model

Procedure Exciton model
for A in $[1,N]$ **do**
  DFT in vacuum; Save density matrix $\mathbf{D}^A$
  ESP fitting in vacuum; Save ESP charges $\mathbf{q}^A$

  TDDFT in vacuum; Solve n states
  Save excitation energies $E_{LE(0)}^{A(n)}$ and CI coefficients $c_{i_A a_A}^{(n)}$

  RODFT for A^+ in vacuum; Save density matrices $\mathbf{D}_\alpha^{A^+}$ and $\mathbf{D}_\beta^{A^+}$
  RODFT for A^- in vacuum; Save density matrices $\mathbf{D}_\alpha^{A^-}$ and $\mathbf{D}_\beta^{A^-}$

for A in $[1,N]$ **do**
  for B in $[A+1,N]$ **do**
    Get density matrix $\mathbf{D}^{AB} = \mathbf{D}^A \oplus \mathbf{D}^B$
    Compute Fock matrix from $\mathbf{D}^{AB}$
    if $N > 2$ **then**
      Add point charge potential to Fock matrix
      $f_{\mu\nu} = f_{\mu\nu} - \sum_{C\neq A,B}\sum_{k\in C} q_k^C \int \mathrm{d}\vec{r}\, \phi_\mu(\vec{r})\, \phi_\nu(\vec{r})(|\vec{r} - \vec{R}_k^C|)^{-1}$

    Transform Fock matrix to MO basis
    Compute 1e terms for LE–CT and CT–CT couplings (Eqs. 12,13,16,17)
    Compute 2e terms for LE–LE, LE–CT and CT–CT couplings (Eqs. 11,12,13,15)

    Get density matrices $\mathbf{D}_\alpha^{A^+B^-}$, $\mathbf{D}_\beta^{A^+B^-}$, $\mathbf{D}_\alpha^{A^-B^+}$, and $\mathbf{D}_\beta^{A^-B^+}$
    Compute CT excitation energies for A^+B^- and A^-B^+ (Eq. 9)

for A in $[1,N]$ **do**
  if $N > 1$ **then**
    Compute point charge potential
    $f_{\mu\nu}^{\Delta} = -\sum_{B\neq A}\sum_{k\in B} q_k^B \int \mathrm{d}\vec{r}\, \phi_\mu(\vec{r})\, \phi_\nu(\vec{r})(|\vec{r} - \vec{R}_k^B|)^{-1}$
    Transform point charge potential to MO basis
    Compute LE excitation energies (Eq. 6)
    Compute LE–LE couplings within A (Eq. 10)
  else
    $E_{LE}^{A(n)} = E_{LE(0)}^{A(n)}$
    $V_{LE-LE}^{A(m),A(n)} = 0$

Diagonalize exciton model Hamiltonian matrix

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we use DFT, TDDFT, and restricted open-shell (RO) DFT, respectively. The partial atomic charges are derived from the ground state electrostatic potential (ESP) for each monomer. In TDDFT calculations, the Tamm−Dancoff approximation (TDA)[48,49] is employed to compute the excitation energies and CI coefficients for singlet excited states of isolated monomers. The second step is to loop over chromophore pairs to calculate the interchromophore LE−LE couplings, as well as energies and couplings involving CT states. In this step, the environmental effects are taken into account by representing the surrounding monomers as ESP-derived point charges. These point charges provide perturbations to the Fock matrix elements through one-electron potential integrals. The third step is to loop over the monomers again to calculate the LE excitation energies and LE−LE couplings within the same monomer, where the environmental effects from the point charges of other chromophores are taken into account. The exciton model Hamiltonian matrix is then constructed and diagonalized to give the eigenvalues and eigenstates, from which the excitation energies for adiabatic excited states can be obtained.

3. RESULTS AND DISCUSSION

Demonstrative calculations using the exciton model were carried out in four systems; (i) dicyanoethylene−furan complex (17 atoms), (ii) zinc-bacteriochlorin (ZnBC)−bacteriochlorin (BC) complex (83 atoms), (iii) bacteriochlorophyll-*a* (BChl-*a*) dimer (152 atoms), and (iv) BChl-*a* assembly (up to 1368 atoms). In the three dimeric systems, full TDDFT calculations and exciton model calculations were performed at a series of separation distances between the chromophore monomers. In full TDDFT calculations for the dimers, we solve for the eight lowest excited states. In the exciton model calculations, we obtain the three lowest excited states for each monomer, together with two intermonomer CT states. Four density functionals were tested in the calculations, including BLYP,[50,51] B3LYP,[52,53] PBE0,[54,55] and LRC-ωPBEh.[23] For the first system, the cc-pVDZ basis set[56] was used, and tensor hypercontraction (THC)[57,58] accelerated EOM-CC2 calculations were carried out in addition to exciton model and TDDFT calculations. For the second system, the LANL2DZ effective core potential basis set[59] was used for the zinc atom and the 6-31G* basis set[60] was used for other atoms. In the third and fourth systems the 6-31G* basis set was used for all atoms. The THC-EOM-CC2 calculations were performed using a development version of the Psi4 program.[61] The CDFT calculations were carried out using the NWChem code.[62] All other calculations were conducted using the GPU-accelerated TeraChem code.[63,64]

3.1. Dicyanoethylene−Furan.

The first system used in exciton model calculations is a molecular pair consisting of furan as the electron donor and dicyanoethylene as the electron acceptor (Figure 1). In this system the molecular planes of the two chromophores were kept parallel, and the vertical distance between these planes was employed as the separation distance *R*. We show in Figure 1 the calculated CT excitation energies with respect to the inverse of *R*, using both the exciton model and full TDDFT calculations with four different density functionals. Because of the small size of the system, EOM-CC2 results are also available and shown as reference.

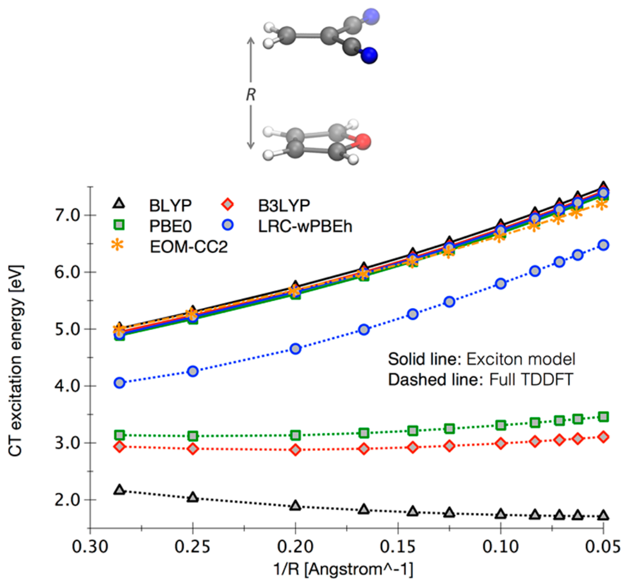

Figure 1. Charge-transfer excitation energies with respect to inverse monomer separation for the dicyanoethylene−furan dimer.

Dreuw and Head-Gordon[20] have shown that TDDFT with the BLYP functional fails to describe the CT excited states. In our calculations, the same conclusion is reflected by the zero asymptotic slope of the dashed black line in Figure 1. The incorrect asymptotic behavior of TD-BLYP results can be traced back to the exchange-correlation kernel used in TDDFT calculations, which decays too quickly with respect to the separation distance between the two chromophores. In addition to the erroneous asymptotic behavior, TD-BLYP calculations also suffer from self-interaction errors in the orbital energies, which leads to significant underestimation of the CT excitation energy. In TDDFT calculations using hybrid density functionals like B3LYP and PBE0, the CT excitation energies become relatively larger, and the asymptotic slope starts to deviate from zero. But the results are still far from satisfactory. Only full TDDFT calculations employing the LRC-ωPBEh functional provide correct asymptotic behavior, but the CT excitation energies still differ from the EOM-CC2 results by around 1 eV over the whole range of separation distances.

In contrast to the largely unsatisfactory and functional-dependent performance of TDDFT, our ab initio exciton model gives robust results for all four density functionals under investigation (solid lines in Figure 1). All the CT excitation energies predicted by the exciton model show correct asymptotic behavior. In addition, the exciton model with different density functionals gives almost identical results for the CT excitation energies, all of which compare well with the EOM-CC2 results over the whole range of separation distances. For this example, our exciton model is robust and outperforms TDDFT with standard parametrizations.

The discrepancy between TD-LRC-ωPBEh and EOM-CC2 results indicates that the standard parameter of $\omega = 0.2$ cannot provide a balanced description for CT excitation in the dicyanoethylene−furan system. In fact, it has been pointed out that the optimal screening parameter ω has a roughly negative correlation with the size of the system under investigation.[65] We therefore tested the LRC-ωPBEh functional with a larger ω value, which is expected to improve the performance of TD-LRC-ωPBEh for this specific case. As shown in Figure 2, the CT excitation energies predicted by TD-LRC-ωPBEh increase significantly as the value of ω increases from 0.2 to 0.4. The asymptotic slope remains constant, whereas the increase in the TDDFT charge-transfer energies arises from the frontier MO energies that are dependent on ω. Among the three tested ω

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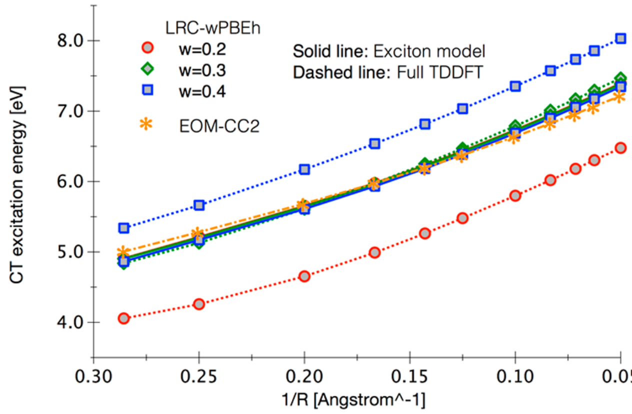

Figure 2. Dependence of charge-transfer excitation energies on distances and ω values for the dicyanoethylene−furan system.

values, 0.3 was found to be optimal, at which the performance of TD-LRC-ωPBEh is comparable with that of EOM-CC2. In contrast to TD-LRC-ωPBEh, the CT energies from the ab initio exciton model are effectively independent of ω, demonstrating its robustness.

It has recently been demonstrated that the screening parameter ω in the LRC-ωPBEh functional can be tuned to give optimal performance for CT excited states, by requiring that the ionization potentials (IP) of the neutral and anionic molecules coincide with the corresponding HOMO energies. Following the IP-tuning procedure[24] we obtained the optimal ω of the dicyanoethylene−furan complex, which was found to be dependent on the separation distance R. The optimal ω, which we denote *IP-tuned* ω, gradually increases as R gets larger, and eventually becomes R-independent when R is greater than 10 Å (Figure 3a, dashed green line). It is interesting to notice that at large R we are actually tuning the IP of the smaller molecule (monomer) since the two molecules are decoupled from each other; whereas at small R the IP of the larger molecule (dimer) is tuned. So the positive correlation between IP-tuned ω and R is in accordance with the previously reported negative correlation with molecular size.[65,66] Since a distance-dependent ω introduces size-inconsistency, we obtained an overall IP-tuned ω of 0.275 by including all the separation distances in the IP-tuning procedure. At this ω the performance of full TDDFT calculations is shown in Figure 3b (dashed green line). Alternatively, we can tune the optimal ω by requiring that the CT excitation energy from the exciton model agrees with that from full TDDFT calculation. Based on the data presented in Figure 2, the dependence of CT excitation energy on ω at each separation distance is interpolated by a quadratic polynomial, and the intersection between the exciton model and full TDDFT corresponds to the optimal ω, which we term as *CT-tuned* ω. The distance-dependence of CT-tuned ω is shown in Figure 3a, where a roughly negative correlation with the separation distance R is observed, opposite to the behavior of the screening parameter in IP-tuned ω. Over the whole separation distance, the CT-tuned ω was obtained as 0.3, and the performance of the exciton model using this ω is shown in Figure 3b (solid blue line). It can be seen that the exciton model with CT-tuned ω agrees nicely with EOM-CC2 at a standard deviation of 0.10 eV, which is smaller than that for full TDDFT calculation with IP-tuned ω (std. dev. = 0.19 eV).

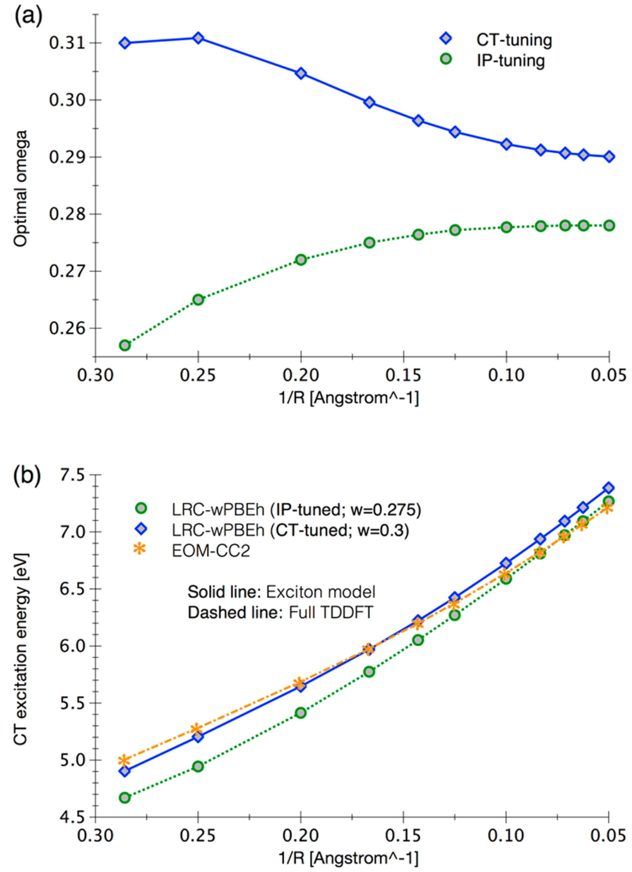

Figure 3. (a) IP-tuned and CT-tuned optimal ω at different separation distances. (b) Comparison between the exciton model with CT-tuned ω and full TDDFT with IP-tuned ω.

The promising performance of the exciton model in reproducing the CT excitation energies relies largely on the evaluation of the diabatic CT energy in the exciton Hamiltonian matrix. Instead of using MO energies and exchange-correlation kernel as in the formulation of TDDFT, we employed a density-matrix-based approach to evaluate the CT energy, as shown in eq 9. As such, the interaction between the hole and the electron is correctly accounted for by the Coulomb term in Kohn−Sham DFT, guaranteeing the correct $1/R$ asymptotic behavior. In addition, the CT excitation energy is evaluated as the energy difference between the ionic and neutral pairs, which is less prone to self-interaction errors or delocalization errors as opposed to the orbital energies.

The idea of minimizing self-interaction-errors for CT excitation energies shares some similarities with the procedure proposed by Dreuw and Head-Gordon[20] but does not require a separate CIS calculation or correction afterward. It is perhaps more appropriate to consider the calculation of CT excitation energies in our exciton model as an approximate CDFT approach,[33] in which localization of densities are enforced through calculations on neutral or ionic monomers. At short separation distance of the dicyanoethylene−furan complex, a comparison between CT excitation energies from our ab initio exciton model and CDFT is shown in Figure S5 in the Supporting Information. At infinite separation, the obtained $E_{CT}^{A\rightarrow B}$ satisfies the requirement that the full-system energy is equal to the sum of the isolated energies.[33] Therefore, the CT excitation energy in the exciton model corrects not only the asymptotic behavior but also the absolute values, making the

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model robust with respect to different density functionals. In addition, we would like to mention the possibility of including more CT states in the exciton model, where the lowest LE state of the dicyanoethylene−furan complex can be further improved (Figure S6).

3.2. Zinc-Bacteriochlorin−Bacteriochlorin. The second system for exciton model calculations is the donor−acceptor complex formed by Zinc-Bacteriochlorin (ZnBC) and bacteriochlorin (BC). The geometries were taken from the literature.[20] This system has been previously used to demonstrate the failure of conventional TDDFT using pure or global hybrid density functionals. We compare exciton model and full TDDFT calculations for this system, plotting the lowest CT excitation energy against the inverse of the center-of-mass distance between ZnBC and BC in Figure 4. Similar to the

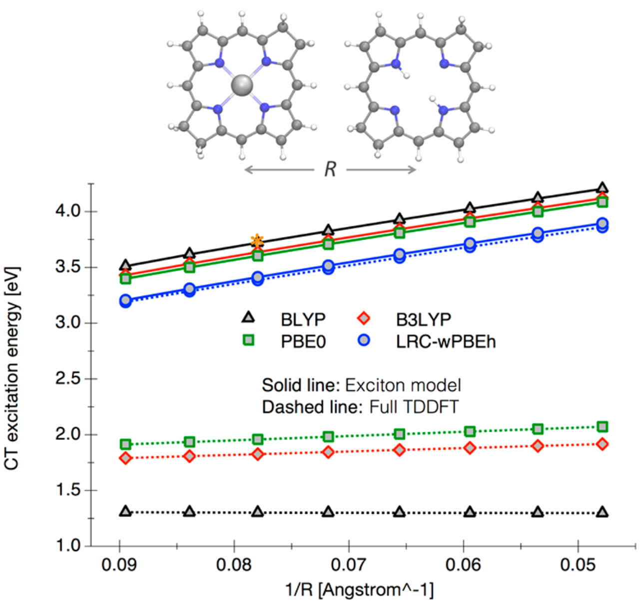

Figure 4. Charge-transfer excitation energies with respect to inverse monomer separation for the ZnBC−BC dimer. The orange asterisk marks the CIS-corrected result of Dreuw and Head-Gordon.[20]

results in Figure 1, TDDFT calculations with BLYP, B3LYP, or PBE0 provide incorrect asymptotic behavior owing to intrinsic deficiencies of the exchange-correlation kernel. The range-separated LRC-ωPBEh functional significantly improves the performance of TDDFT for CT excited states, providing correct asymptotic behavior and CT excitation energies comparable to previous reports.[20,35]

In contrast to TDDFT, our exciton model gives correct asymptotic behavior and robust CT excitation energies for all the density functionals under investigation. Only a small dependence on the density functional was observed (~0.3 eV difference between BLYP and LRC-ωPBEh, as opposed to ~2.0 eV for full TDDFT calculations). Moreover, the CT excitation energy obtained with BLYP is in good agreement with previous reports using the same functional and a CIS-based correction method (orange asterisk in Figure 4). At the same geometry of the ZnBC−BC complex used in previous studies,[20,35] we obtained a ZnBC → BC charge-transfer energy of 3.72 eV, which compares well with 3.75 eV obtained by CIS-based correction[20] and 3.79 eV from CDFT.[35] It is also noteworthy that the results from the exciton model and TDDFT calculations almost coincide with each other when the LRC-ωPBEh functional is used. We also compare the performance of our exciton model and TDDFT with a tuned screening parameter ω. The IP-tuned and the CT-tuned optimal ω values are shown in Figure 5a. It can be seen that the IP-tuned ω

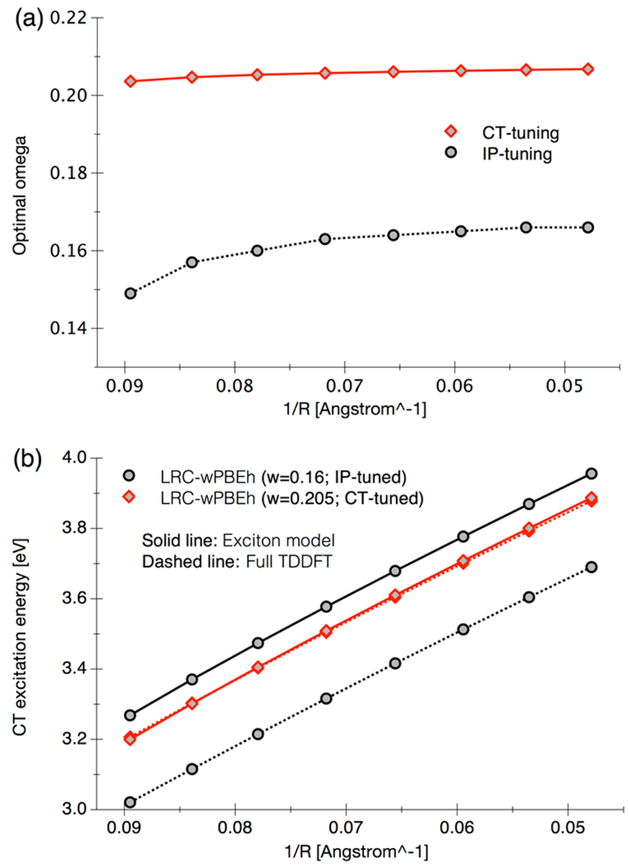

Figure 5. (a) IP-tuned and CT-tuned optimal ω at different separation distances for the ZnBC−BC complex. (b) Comparison between the exciton model and full TDDFT at optimal ω values.

increases as the separation distance R increases, whereas the CT-tuned ω exhibits almost negligible dependence on R. The comparison between exciton model and full TDDFT calculations with optimal ω values is shown in Figure 5b. At the CT-tuned optimal ω (0.205), the results from the exciton model and full TDDFT almost coincide (solid and dashed red lines), whereas at the IP-tuned optimal ω (0.16), the CT energy from full TDDFT (dashed black line) is ~0.25 eV smaller than that from the exciton model (solid black line). Unfortunately, higher level calculations such as EOM-CC2 for the ZnBC−BC complex are not presently available. Considering the fact that the exciton model agrees excellently with previous reports when using the BLYP functional (Figure 4), we infer that the exciton model with CT-tuned ω performs better than full TDDFT with IP-tuned ω.

3.3. Bacteriochlorophyll-*a* Dimer. The third system used in exciton model calculations is the BChl-*a* dimer. The choice of this system is motivated by the fact that the BChl-*a* chromophore constitutes the B800 and B850 antenna in the LH2 light-harvesting complex, and we would like to use the exciton model to calculate excitation energies of similar multichromophore complexes. The geometry of the dimer was taken from the crystal structure of the LH2 complex from *Rps. acidophila*.[67] The excitation energy of the lowest CT state

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is plotted against the center-of-mass distance between the two BChl-*a* chromophores, as shown in Figure 6. Again, the exciton

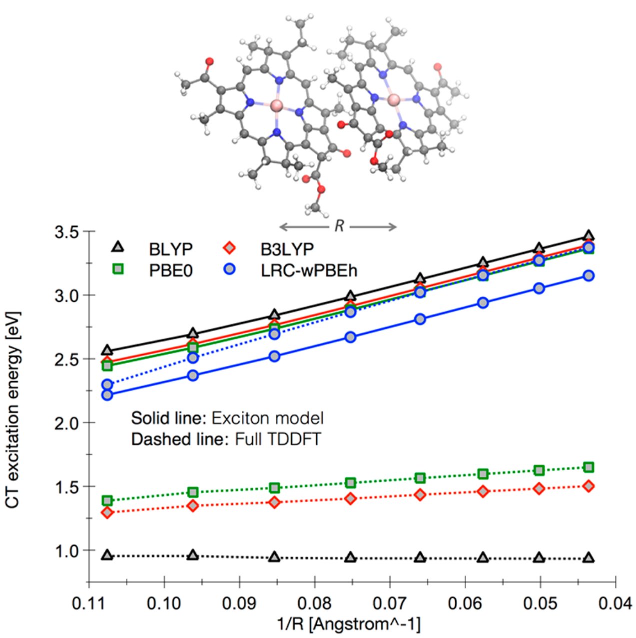

Figure 6. Charge-transfer excitation energies with respect to the inverse separation distance (between monomer centers of mass) for the BChl-*a* dimer.

model outperforms TDDFT calculations with either pure or global hybrid density functionals. The curves calculated by the exciton model using different functionals are parallel to each other, confirming the robustness of the exciton model.

Interestingly, we observed that the performance of TD-LRC-ωPBEh calculations exhibits an unusual dependence on the distance between the chromophore monomers, where the CT excitation energy decreases quickly as the two chromophores approach closer than 10 Å (Figure 6, dashed blue line). This is most likely due to the screening parameter ω, which introduces distance dependence to the fraction of Hartree−Fock exchange. At large separation, the amount of Hartree−Fock exchange approaches 100%, ensuring the correct $1/R$ asymptotic behavior of the CT excitation energy. At shorter distance, the Hartree−Fock exchange becomes screened, and the faster decay of CT excitation energy with respect to $1/R$ can be attributed to the narrowed orbital gap. This implies that optimal description of valence and CT excitations with LRC functionals within TDDFT may require different ω values, depending on molecular size and charge-transfer distance. Our exciton model removes this sensitivity.

To test the dependence on the screening parameter, we carried out exciton model and full TDDFT calculations on the BChl-*a* dimer, using the LRC-ωPBEh functional with different ω values. In addition to the standard value of 0.2, two smaller ω values of 0.15 and 0.11 were tested, as shown in Figure 7. As also observed in the ZnBC−BC complex, as ω gets smaller, the exciton model and full TDDFT calculations show opposite trends in CT excitation energies. The behavior of the exciton model is rather robust, with changes in the CT energy being restricted to 0.2 eV for these variations of ω. In contrast, the ω-dependence of the CT energy from TD-LRC-ωPBEh calculations is quite significant (0.4−0.6 eV). All of the excitation energies (from both exciton and LRC-TDDFT

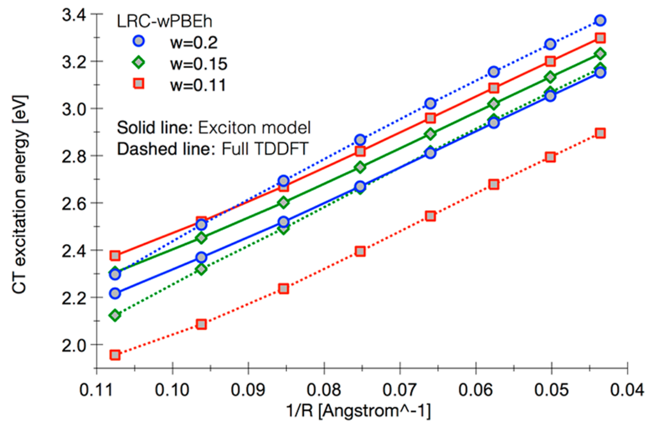

Figure 7. Dependence of charge-transfer excitation energies on distances and ω values for BChl-*a* dimer.

models) are parallel at large distance, but the LRC-TDDFT results are no longer parallel to the exciton model results at short distances.

To further understand the effects of the screening parameter ω, we plotted the absorption spectrum of the BChl-*a* dimer at the shortest separation distance (Figure 8), which is the same as that in the crystal structure. The absorption spectra were obtained by Gaussian broadening with a full-width at half-maximum of 0.05 eV. The maximal absorption wavelength becomes shorter as ω gets smaller, and this trend is systematic in the absorption spectra calculated using the exciton model. However, the behavior of the spectra from TDDFT is less systematic. At the smallest ω of 0.11, the absorption spectrum from TD-LRC-ωPBEh shows a spurious band at 625 nm, which does not appear in the calculations using larger ω. To verify the origin of this spurious band, we carried out another exciton model calculation in which the diabatic CT excitation energies are determined by the TDDFT-TDA formalism, i.e., from eq 8 instead of eq 9. The result suggests that the absorption band at 625 nm corresponds to a mixed LE/CT state with a significant portion of CT transitions (see the Supporting Information for details). The origin of such mixing lies in the different behavior of LE and CT excitation energies with respect to ω values. In TDDFT-TDA, the former increases whereas the latter decreases as ω gets smaller, leading to excessive mixing between LE and CT states at small ω. In our exciton model where the CT energies are calculated from density matrices as in eq 9, the LE and CT energies show similar trends as ω is varied, and the results are thus more robust.

It is also of interest to compare the IP-tuned and CT-tuned ω values, as shown in Figure 9. The CT-tuned ω varies monotonically between 0.19 and 0.16 as the separation distance increases, whereas the IP-tuned ω has smaller magnitude between 0.09 and 0.14. We have shown in Figure 8 that TDDFT calculation with an ω value of 0.15 gives rise to a weak absorption band at around 575 nm due to underestimation of CT excitation energies. The IP-tuned ω, which is even smaller, will not be able to properly describe the interplay between LE and CT states in TDDFT calculations. The reason is that the BChl-*a* dimer has less CT character than the other dimeric systems studied in this work, and the HOMOs of the neutral and the anionic system are mostly located on the same BChl-*a* chromophore. This means that the IP-tuning procedure is effectively optimizing ω for a monomer instead of a dimer. It is therefore not surprising that the IP-tuned ω fails to describe

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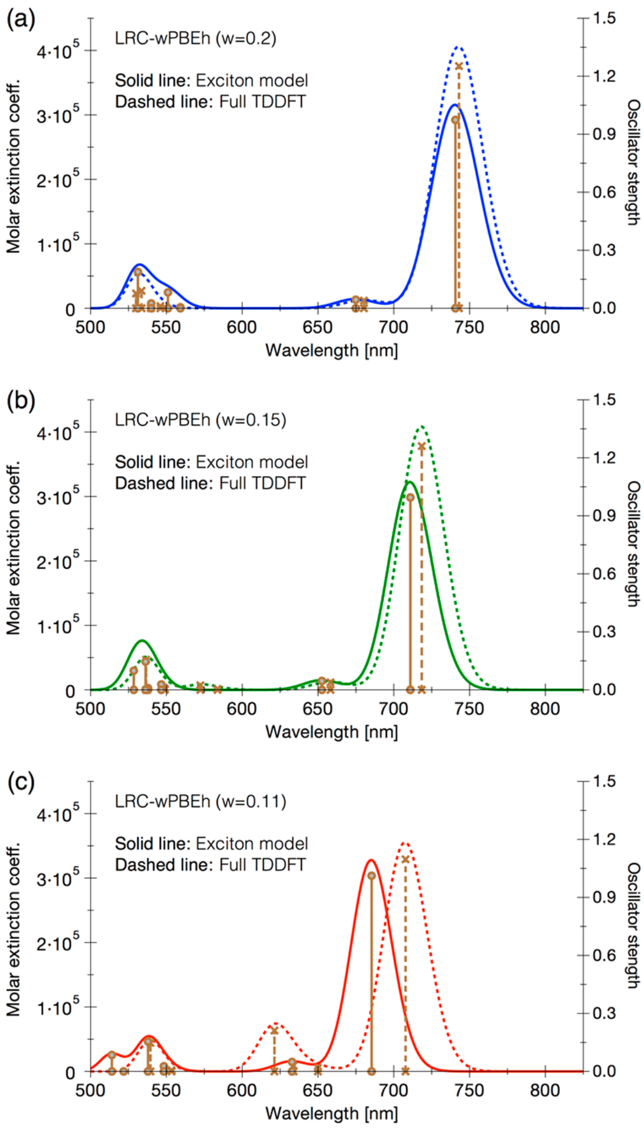

Figure 8. Dependence of absorption spectra on ω values for BChl-*a* dimer.

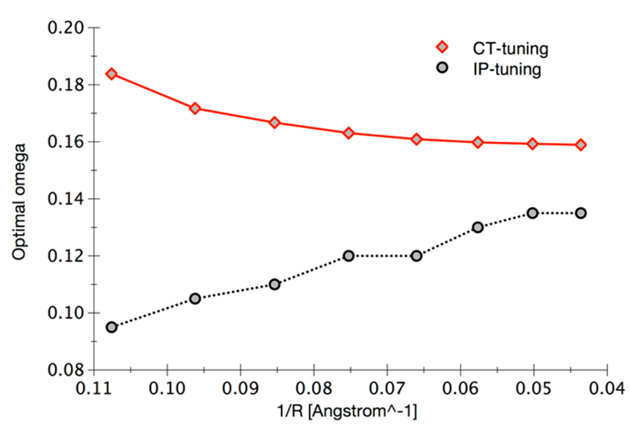

Figure 9. Optimal ω values for the BChl-*a* dimer at different separation distances.

CT within the BChl-*a* dimer. The CT-tuning procedure, instead, takes into account both chromophore monomers, and thus provides more reliable results compared to IP-tuning. The CT-tuned ω over all separation distances is obtained as 0.165.

3.4. Bacteriochlorophyll-*a* Assembly. Using the CT-tuned ω of 0.165 for BChl-*a* dimer, we proceeded to apply the exciton model to the whole B850 assembly in the LH2 B800−B850 complex, which consists of 18 BChl-*a* chromophores. We consider contiguous subsets of the BChl-*a* chromophores, ranging from a single BChl-*a* chromophore to the entire 18-chromophore B850 assembly. The absorption spectra are calculated from the crystal structure and do not involve conformational disorder, apart from the Gaussian broadening. As the number of the BChl-*a* chromophores increases, the absorption spectrum (Figure 10) shows a considerable red-shift

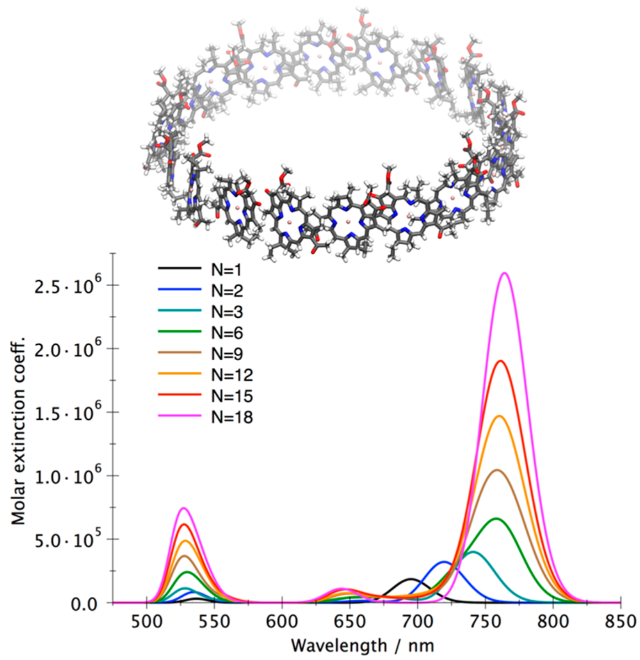

Figure 10. Absorption spectra of BChl-*a* assemblies including different number of chromophores.

as well as an increase in the molar extinction coefficient. The lowest excitation energy (S_1) for the whole B850 assembly is calculated as 1.60 eV, which is close to the experimental value of ∼1.45 eV.[68] The experimental value corresponds to the entire B800−B850 complex in the protein environment. Our model includes only the B850 assembly and excludes the B800 assembly, carotenoids, and surrounding protein. Thus, we do not expect to exactly reproduce the experimental value, which is only provided for reference. Shown in Figure 11 is a further comparison among the lowest excitation energies from the Frenkel exciton model, the exciton model with CT states, and the full TDDFT calculations. It can be seen that inclusion of the CT states in the exciton model provides a red-shift of 0.08 eV in the S_1 excitation energy compared with the Frenkel exciton model. Full TDDFT calculations red-shift the S_1 excitation energy even further to 1.53 eV.

Finally we compare the computational efficiency of the exciton model and full TDDFT. The timings of calculations for B850 assemblies are shown in Figure 12, obtained on a desktop workstation using dual Intel Xeon X5680 CPUs, 144 GB RAM, and 8 GeForce GTX 970 GPUs. As the size of the system increases, the exciton model shows more favorable scaling with respect to system size of $O(N^{1.4})$ compared to $O(N^{1.9})$ for full TDDFT calculations. Moreover, the prefactors are comparable, so the exciton model already outperforms full TDDFT

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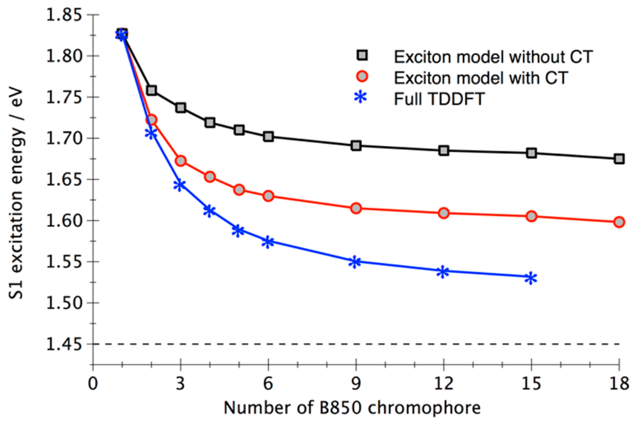

Figure 11. Comparison between the lowest excitation energy from the Frenkel exciton model, the exciton model with CT states, and full TDDFT calculations. The dashed line denotes the experimental excitation energy of 1.45 eV for the B800–B850 complex in LH2 (see text).

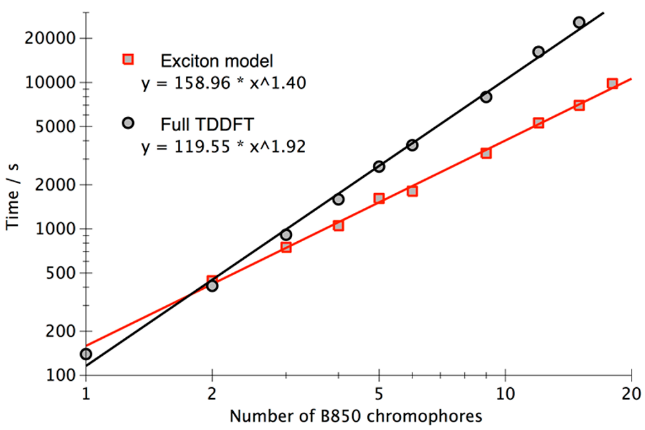

Figure 12. Comparison between the computational cost of the exciton model and full TDDFT calculations.

calculations once the complex has at least three chromophores. It should also be pointed out that the full TDDFT calculations in Figure 12 compute only the four lowest states. In contrast, the number of excited states computed in the exciton model increases quadratically with respect to the number of chromophores, up to 360 excited states (54 LE states + 306 CT states) for the whole B850 assembly. Solving for all of these states using full TDDFT would be very time-consuming, requiring more Davidson–Liu iterations.[69] Therefore, the exciton model scales considerably better than full TDDFT calculations. The computational cost of the exciton model may be further decreased by exploring hierarchical parallelization as well as numerical approximations for distant chromophores.

4. CONCLUSIONS

In summary, we have presented an expanded exciton model that includes both valence and charge-transfer excited states in the Hamiltonian matrix. The performance of the exciton model is promising in that it provides not only correct asymptotic behavior but also reliable charge-transfer excitation energies. Demonstrative calculations in three dimeric systems and one multichromophoric system confirm that the exciton model is robust with respect to the chosen density functional as well as the screening parameter in long-range corrected hybrids. The exciton model is also able to guide the choice of the screening parameter for a specific system, where the standard or the IP-tuned parameters may not be optimal. The exciton model developed in this work scales better than full TDDFT calculations, is readily applicable to systems consisting of a large number of chromophores, and serves as a starting point for future studies involving more charge-transfer states as well as double-exciton states.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.7b00171.

Details concerning ESP fitting used in the extended exciton model. Analysis of size-extensivity in the exciton model. Effects of CT states and interchromophore overlap on exciton energetics. Comparison between exciton model and CDFT. (PDF)

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Notes

The authors declare the following competing financial interest(s): T.J.M. is a co-founder of PetaChem, LLC.

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